Meeting Summaries of April 30

Date: April 30
Time: Not recorded
Attendees: Dr. Jabed and Feng
Topic: Review of chemical data quality issues, unit inconsistencies in mercury and zinc measurements, reproduction of prior ordination results, and next steps for data cleaning and conference preparation.

Questions, Feedbacks and Suggestions

- Dr. Jabed: The urgent issue is to **resolve the data-unit inconsistencies immediately**. Feng should contact Jian as soon as possible, since Jian was involved in the original data preparation and can clarify the intended measurement units and scaling choices.

  > Key point for the next meeting
  >
  > Confirm the correct units and scaling for the chemical dataset first. The analysis should not move forward until the **original units** in the dataset and the **practical units** used in analysis are clearly resolved.

- Dr. Jabed: If Jian cannot be reached quickly, Feng should also contact Prof. Jan Siborowski within the following days to discuss the original sampling and data handling context.

- Dr. Jabed: If this unit issue can not be solved, Feng should email the relevant **chapter from Jian's thesis** to Dr. Jabed and Dr. Yue so the team can compare the current workflow with the original study and identify where the data handling may differ.

- Dr. Jabed: The current dataset should be checked for **unit consistency across years**, the mercury and zinc measurements from 1999 compared with later records from 2004–2005 in the Detroit River data.

- Dr. Jabed: If the data issues are not resolved quickly, the team may need to **review Jian's thesis chapter directly** to provide more specific guidance and avoid repeated rework.

- Dr. Jabed: Feng should continue preparing for the upcoming **conference presentation**, showing the latest progress without the final results is acceptable.

Summary of Discussion

Chemical Data Quality and Unit Inconsistencies

The meeting focused on persistent data quality issues in Feng's chemical dataset, particularly inconsistencies in measurement units across different years. Feng explained that **mercury** and **zinc** values appeared especially problematic, with some extreme observations from 1999 differing sharply from later Detroit River records collected in 2004–2005. These differences raised concerns that some values may have been entered under different units or scaled incorrectly.

Dataset Structure and Anomaly Detection

Feng reported that the working chemical dataset contains **322 total records**, formed by combining two source files with 245 and 77 sites, respectively. The dataset includes 16 chemical variables and 6 sample information fields such as site identifiers, years, and water body metadata. During screening, Feng identified suspicious mercury and zinc values.

Results Comparison and Reproduction of Earlier Work

Feng presented updated results after correcting the most obvious unit problems. These revisions improved the overall distribution of the contamination scores and produced patterns that looked more plausible than the earlier flattened results. However, Feng also noted that the current analysis still does not reproduce exactly the same loading structure reported in the original 2008 study. While most variables now show similar relative magnitudes, some differences remain, particularly for copper and for the relationship between certain organic contaminants. The meeting concluded that exact reproduction cannot be evaluated fairly until the unit problem is settled with confidence.

Data Cleaning Priorities and Escalation Plan

Dr. Jabed advised Feng to contact Jian and arrange a discussion within the following days. Because Jian and Prof. Jan Siborowski were involved in the original data collection and processing, they are the most direct source for clarifying whether the problematic values reflect unit differences, transcription errors, or other historical data issues. Dr. Jabed also asked Feng to share the relevant chapter from Jian's thesis with Dr. Jabed and Dr. Yue. If the matter is not resolved quickly, the team will review that chapter themselves to identify the intended workflow and provide additional guidance.

Conference Preparation and Follow-Up

The urgency of the data issue was linked to Feng's upcoming conference presentation in late May. Dr. Jabed advised Feng to continue preparing the presentation using the best current results, while clearly stating that parts of the chemical-data analysis are still being cleaned and verified. The team agreed to hold a follow-up meeting next Thursday at 7:00 PM to review any updates from Jian and Prof. Jan Siborowski and to decide whether further changes to the analysis strategy are needed.

Action Items

- Email **Jian** to explain the unit inconsistencies and arrange a meeting as quickly as possible, ideally within the next 1–2 days.

- If Jian is unavailable, contact **Prof. Jan Siborowski** to discuss the original data collection and the likely source of the unit inconsistencies.
- Email the relevant **chapter from Jian's thesis** to Dr. Jabed and Dr. Yue for reference.
- Continue **checking and cleaning the dataset**, with particular attention to mercury and zinc values, unit consistency, and scaling across years.
- Verify the two chemical source files containing **245 and 77 sites**, respectively, before merging them and treating the combined dataset as final.
- Prepare for the **late-May conference presentation**, using the best current results while noting clearly which analyses are still in progress.
- Attend the **follow-up team meeting next Thursday at 7:00 PM**.

Follow-Up

Next meeting time: May 7, 19:00 (local time)

Supportive Materials

- A meeting recording was saved on Feng's local drive.